

Course Name: CSA16 - Data Warehousing and Data Mining	Course Outcome: CO2 (BL3, BL4)
Weightage / Total Marks: 50% 50 Marks	Assessment Type: Scenario-Based Questions

Q1. Data Preprocessing in Retail Analytics (10 Marks)

A. Handling Missing Values

i. Identification of Missing Values:

- Age is missing in record **C2** (NaN).
- Monthly_Spend is missing in record **C3** (NaN).
- Purchase_Frequency is missing in record **C5** (NaN).

ii & iii. Imputation Techniques, Calculations, and Justification:

• Method 1: Median Imputation (Robust to Outliers)

Computation: Valid Age values are {23, 31, 105, 27, 40, 36, 42}. Sorting valid values gives {23, 27, 31, 36, 40, 42, 105}. The median is **36**.

Imputed Value for C2 (Age): **36 years**.

Justification: Because Age contains an extreme outlier (105), median imputation is preferred over mean imputation as it is non-parametric and unaffected by extreme skewed values.

• Method 2: k-Nearest Neighbors (k-NN) / Group-Based Imputation

Computation for Monthly_Spend (C3 - Female, High Loyalty): High loyalty customers (C4, C8) have spends of ₹500K and ₹300K (or average spend among non-outlier high loyalty). If using median of valid non-outlier spends {20, 25, 35, 38, 45, 300, 500}, global median = **38**.

Computation for Purchase_Frequency (C5 - Male, Low Loyalty, Spend ₹25K): Looking at similar Low loyalty records (C1: Spend 20, Freq 5; C7: Spend 38, Freq 9), k-NN (k=2) calculates mean frequency = $(5 + 9) / 2 = 7$.

Imputed Values: Monthly_Spend (C3) = **₹38K** (Global Median) or **₹45K** (Group Median); Purchase_Frequency (C5) = **7**.

Justification: k-NN preserves feature correlations across multiple dimensions (Gender, Loyalty, Spend) far better than single-variable imputation.

B. Outlier Detection and Treatment

i. Z-Score Method Formula: $Z = (X - \mu) / \sigma$

Using population parameters for raw numeric data:

- Age: Mean $\mu = 43.43$, Std Dev $\sigma = 25.92$
- Monthly_Spend: Mean $\mu = 138.29$, Std Dev $\sigma = 174.19$

ii. Identification of Extreme Values:

- Customer C4 (Age = 105): $Z = (105 - 43.43) / 25.92 = +2.38$. An age of 105 is biologically unrealistic for typical retail customer profiles and represents a data entry error.
- Customer C4 (Spend = ₹500K): $Z = (500 - 138.29) / 174.19 = +2.08$. Extremely high spend relative to standard retail transactions.
- Customer C8 (Spend = ₹300K): High value customer ($Z = +0.93$), valid high-tier buyer.

iii. Treatment Methods and Justification:

- **Capping / Winsorization:** Cap Age at an upper threshold of 65 or 99th percentile (e.g., set Age 105 → 65). Cap extreme Monthly_Spend at 95th percentile (e.g., ₹150K).
- **Removal:** If C4 represents a systemic logging error across all attributes, remove record C4.
- *Justification:* Capping prevents loss of valuable customer records while stopping extreme skewness from distorting distance-based algorithms in recommendation systems.

C. Categorical Data Standardization

i. **Inconsistencies Identified:** Variations in casing, abbreviations, and formatting for Gender (M, Male, male, F, Female, FEMALE).

ii. Standardized Mapping Table:

Customer_ID	Raw Gender	Standardized Gender	Standardization Rule Applied
C1	M	Male	Map 'M' → 'Male'
C2	Male	Male	Retain standard casing
C3	F	Female	Map 'F' → 'Female'
C4	Female	Female	Retain standard casing
C5	male	Male	Title Case conversion
C6	FEMALE	Female	Title Case conversion
C7	M	Male	Map 'M' → 'Male'
C8	Female	Female	Retain standard casing

Q2. Covariance & Data Reduction in E-commerce Analytics (10 Marks)

A. Covariance Analysis

i. Covariance Computation between Income (X) and Purchase_Value (Y):

Formula: $Cov(X, Y) = \Sigma [(X_i - \bar{X})(Y_i - \bar{Y})] / (n - 1)$

- $\bar{X}$ (Mean Income) = $649 / 8 = 81.125$ ₹K
- $\bar{Y}$ (Mean Purchase Value) = $1590 / 8 = 198.75$ ₹K

Cust	X (Income)	Y (Purchase)	(X - X̄)	(Y - Ȳ)	(X - X̄)(Y - Ȳ)
U1	40	100	-41.125	-98.75	4061.09
U2	60	150	-21.125	-48.75	1029.84
U3	85	210	3.875	11.25	43.59
U4	120	320	38.875	121.25	4713.59
U5	45	110	-36.125	-88.75	3206.09
U6	75	180	-6.125	-18.75	114.84
U7	150	360	68.875	161.25	11106.09
U8	70	160	-11.125	-38.75	431.09
Sum Σ [(X - X̄)(Y - Ȳ)]					24,706.25

$$Cov(Income, Purchase_Value) = 24,706.25 / (8 - 1) = +3529.46 \text{ (Sample Covariance)}$$
$$Cov_pop(Income, Purchase_Value) = 24,706.25 / 8 = +3088.28 \text{ (Population Covariance)}$$

ii. **Interpretation:** The covariance value is **positive (+3529.46)**. This indicates a strong direct linear relationship: as a customer's income increases, their purchase value also increases proportionally.

iii. **Role of Covariance in Feature Relationships:** Covariance indicates the directional linear dependency between continuous attributes. It allows data engineers to detect multicollinearity and feature redundancy, enabling elimination of duplicate feature variance.

B. Feature Selection

i. **Identification of Redundant Attributes:**

- Income, Purchase_Value, Avg_Balance, and EMI exhibit near-perfect positive correlation ($r \approx 0.98 - 0.99$).
- Orders scales directly with Purchase_Value.

ii. **Optimal Feature Subset:** Select {Income, Credit_Score, Orders}.

iii. **Justification:**

- Income captures customer financial capacity (eliminates redundant Purchase_Value, Avg_Balance, EMI).
- Credit_Score captures risk level and financial trustworthiness (uncorrelated behavioral dimension).
- Orders captures engagement frequency. This set minimizes collinearity while retaining maximum variance.

C. Data Transformation

i. **Standardization (Z-Score) Table (excluding Cust_ID & Customer_Class):**

Formula: $Z = (X - \mu) / \sigma$ using Population Standard Deviation (σ_pop):

Cust	Age (Z)	Income (Z)	Purchase (Z)	Credit (Z)	Orders (Z)	Avg_Bal (Z)	EMI (Z)
U1	-1.50	-1.16	-1.12	-1.35	-1.27	-1.06	-1.04
U2	-0.53	-0.59	-0.55	-0.43	-0.65	-0.58	-0.58
U3	0.45	0.12	0.13	-0.05	0.34	-0.02	-0.12
U4	1.23	1.12	1.37	1.43	1.22	1.18	1.22
U5	-1.01	-1.01	-1.01	-0.79	-0.90	-0.90	-0.93
U6	-0.04	-0.16	-0.21	-0.24	-0.15	-0.26	-0.25
U7	1.62	1.97	1.83	1.80	1.85	1.98	1.89
U8	-0.23	-0.30	-0.44	-0.33	-0.40	-0.42	-0.36

ii. Why Normalization/Standardization is Required: Features measured in different scales (e.g., Age in tens vs EMI in thousands) cause distance-based machine learning algorithms (k-NN, SVM, Gradient Descent) to be dominated by high-magnitude variables. Normalization transforms all features to a common scale, ensuring balanced weight distribution and fast numerical convergence.

Q3. Data Transformation & Discretization in Education Analytics (10 Marks)

A. Equal-Width Binning

Formula: $Width = (Max - Min) / N$ (where $N = 3$)

- **Study_Hours:** Min = 2, Max = 11. $Width = (11 - 2) / 3 = 3.0$.
 Bin 1: [2.0, 5.0] → {S1(2), S2(3), S3(4), S4(5)}
 Bin 2: (5.0, 8.0] → {S5(6), S6(7), S7(8)}
 Bin 3: (8.0, 11.0] → {S8(9), S9(10), S10(11)}
- **Test_Score:** Min = 40, Max = 100. $Width = (100 - 40) / 3 = 20.0$.
 Bin 1: [40.0, 60.0] → {S1(40), S2(50), S3(55)}
 Bin 2: (60.0, 80.0] → {S4(65), S5(75)}
 Bin 3: (80.0, 100.0] → {S6(85), S7(90), S8(95), S9(98), S10(100)}

B. Equal-Frequency Binning

Total Records = 10. For $N = 3$ bins, target count per bin ≈ 3 or 4 records (3, 3, 4 distribution):

- **Study_Hours:**
 Bin 1 (3 records): {S1(2), S2(3), S3(4)}
 Bin 2 (3 records): {S4(5), S5(6), S6(7)}
 Bin 3 (4 records): {S7(8), S8(9), S9(10), S10(11)}
- **Test_Score:**
 Bin 1 (3 records): {S1(40), S2(50), S3(55)}
 Bin 2 (3 records): {S4(65), S5(75), S6(85)}
 Bin 3 (4 records): {S7(90), S8(95), S9(98), S10(100)}

C. Concept Hierarchy Construction

Mapping continuous variables into categorical semantic abstraction levels:

- **Study_Hours Concept Hierarchy:**

- Low: [2, 5] hours (S1 - S4)
- Medium: [6, 8] hours (S5 - S7)
- High: [9, 11] hours (S8 - S10)

- **Test_Score Concept Hierarchy:**

- Poor: [0, 50] marks (S1, S2)
- Average: [51, 75] marks (S3, S4, S5)
- Excellent: [76, 100] marks (S6 - S10)

Q4. Normalization of Numeric Data Array (10 Marks)

Given Data Array: {200, 300, 400, 600, 1000}

(a) Min-Max Normalization (Target Range [0, 1])

Formula: $v' = (v - \text{Min}) / (\text{Max} - \text{Min})$ where $\text{Min} = 200$, $\text{Max} = 1000$, $\text{Max} - \text{Min} = 800$

- For $v = 200$: $v' = (200 - 200) / 800 = 0.000$
- For $v = 300$: $v' = (300 - 200) / 800 = 100 / 800 = 0.125$
- For $v = 400$: $v' = (400 - 200) / 800 = 200 / 800 = 0.250$
- For $v = 600$: $v' = (600 - 200) / 800 = 400 / 800 = 0.500$
- For $v = 1000$: $v' = (1000 - 200) / 800 = 800 / 800 = 1.000$

Normalized Array: {0.000, 0.125, 0.250, 0.500, 1.000}

(b) Z-Score Normalization

Formula: $v' = (v - \text{Mean}) / \text{StdDev}$

- **Mean (μ)** = $(200 + 300 + 400 + 600 + 1000) / 5 = 2500 / 5 = 500$
- **Variance (σ^2)** = $[(200-500)^2 + (300-500)^2 + (400-500)^2 + (600-500)^2 + (1000-500)^2] / 5$
= $[90000 + 40000 + 10000 + 10000 + 250000] / 5 = 400000 / 5 = 80000$
- **Population Std Dev (σ)** = $\sqrt{80000} \approx 282.843$ (Sample Std Dev $s \approx 316.228$)

Calculated Z-Scores (using Population Std Dev):

- For $v = 200$: $v' = (200 - 500) / 282.843 = -300 / 282.843 = -1.061$
- For $v = 300$: $v' = (300 - 500) / 282.843 = -200 / 282.843 = -0.707$
- For $v = 400$: $v' = (400 - 500) / 282.843 = -100 / 282.843 = -0.354$
- For $v = 600$: $v' = (600 - 500) / 282.843 = +100 / 282.843 = +0.354$
- For $v = 1000$: $v' = (1000 - 500) / 282.843 = +500 / 282.843 = +1.768$

Normalized Array: {-1.061, -0.707, -0.354, +0.354, +1.768}

Q5. Data Transformation in Online Shopping Behavior Analysis (10 Marks)

A. Equal-Frequency Binning

i & ii. Division into 3 Bins (10 records total: 3, 3, 4 distribution):

- **Bin 1 (Short Visits - 3 records):** C1 (15s), C2 (30s), C3 (45s) → Range: [15s, 45s]
- **Bin 2 (Moderate Visits - 3 records):** C4 (60s), C5 (120s), C6 (300s) → Range: (45s, 300s]
- **Bin 3 (Long Visits - 4 records):** C7 (600s), C8 (900s), C9 (1800s), C10 (3600s) → Range: (300s, 3600s]

iii. **Analysis of Grouping Effectiveness:** Equal-frequency binning handles highly skewed datasets (like visit duration) exceptionally well because it guarantees balanced cluster sizes. Short visits are cleanly separated from power users, preventing high-frequency small values from dominating a single bucket.

B. Z-Score Normalization

i. Mean and Standard Deviation Computation:

- *Mean (μ) = (15 + 30 + 45 + 60 + 120 + 300 + 600 + 900 + 1800 + 3600) / 10 = 747.0 sec*
- *Population Std Dev (σ) = 1090.92 sec (Sample Std Dev s = 1149.93 sec)*

ii. Computation Table:

Customer_ID	Visit Duration (s)	Z-Score Formula (X - 747) / 1090.92	Z-Score Result
C1	15	(15 - 747) / 1090.92	-0.671
C2	30	(30 - 747) / 1090.92	-0.657
C3	45	(45 - 747) / 1090.92	-0.643
C4	60	(60 - 747) / 1090.92	-0.630
C5	120	(120 - 747) / 1090.92	-0.575
C6	300	(300 - 747) / 1090.92	-0.410
C7	600	(600 - 747) / 1090.92	-0.135
C8	900	(900 - 747) / 1090.92	+0.140
C9	1800	(1800 - 747) / 1090.92	+0.965
C10	3600	(3600 - 747) / 1090.92	+2.615

iii. **Extreme Value Handling:** Z-score normalization preserves outlier information without compressing all non-outlier data into an unresolvable tiny margin. Customer C10 receives a score of **+2.615**, signaling to clustering algorithms that it is a statistical outlier while keeping standard visits smoothly scaled around negative values.

C. Min-Max Normalization

i & ii. Min-Max Transformation [0, 1]:

Formula: $X' = (X - 15) / (3600 - 15) = (X - 15) / 3585$

Customer_ID	Visit Duration (s)	Min-Max Calculation	Normalized Value
C1	15	(15 - 15) / 3585	0.0000
C2	30	(15 / 3585)	0.0042
C3	45	(30 / 3585)	0.0084
C4	60	(45 / 3585)	0.0126
C5	120	(105 / 3585)	0.0293
C6	300	(285 / 3585)	0.0795
C7	600	(585 / 3585)	0.1632
C8	900	(885 / 3585)	0.2469
C9	1800	(1785 / 3585)	0.4979
C10	3600	(3585 / 3585)	1.0000

iii. Comparison of Scaling & Sensitivity to Outliers:

Method Comparison Summary:

- **Min-Max Sensitivity:** Min-Max normalization is *highly sensitive to extreme outliers*. The maximum value (3600s) forces 60% of the dataset (C1 through C6) to be squished into a tiny range between 0.0000 and 0.0795, squashing variance among short duration users.
- **Z-Score Superiority for Clustering:** Z-Score normalization does not bound values to a fixed interval, preserving the relative distance variance across small values while placing outliers at an appropriate distance scale ($Z > +2.5$). Therefore, Z-Score normalization is far better suited for distance-based clustering algorithms like K-Means.